

4 Dars.

Taqrorlash.

Alignment LLM larni foydalar zararsiz va xq/ol
bo'lishga o'rgatish.

Attention → Modelga kiritilgan harflar pozitsiyalariga
selectiv e'tibor berish imkonini beruvchi mexanizm.

BPE → Byte Pair encoding GPT va Llama
tokenization algoritmi.

Chipdilla Law Berilgan compute byudjeti
uchun optimal to'la soni parametrlarining 20
barobarini

Common Crawl → LLM pre training uchun ishlatiladigan
katta web crawl: filte talab qilish.

Deduplication → Ta'ror malumotini olib tashlash.

Emergence Oqibatlash mashtabini to'g'ri
paydo bo'lishi asta sekinlash.

Flop compute narxining standart birligi.

RLHF alignment texnikasi.

Scaling Law Model Natijasi mashtab o'zgarishi
qimmat qonuni.

1. Parametr 12. L. D² model parametrlari
sonini taxminiy hisoblash, L-laser, D-toray
2. Kotiza sigimi 16 B x N
3. Complete by budget GNP jami o'zlash
4. Vaght. GNP / (GPU.perc. MFU)
5. Narx Soat. GPU. \$ / soat
6. Sarafdorlik MFU · BF16 · checkpoint

Embedding.

Söz, token, rasmiy duoz, kabi diskret kate-
gorik dasturlarni past o'lchamli rangarlik
vektorlarga afflaytirish. Asosiy LLM no text yas
math. model matn bilan emous fagat
matematika bilan ishlaydi

Matn fagat inson model interfeysi
Hello sozi [0, 23, -0, 49, 0, 23, ...]
kabi floating point vektorlarga aylantiriladi.
O'pkhon modelga qarab farqlanadi.
Llama 4096, Gemini 3072

Vektor o'zaro idagi masofa bor. Maso jikabdi
yag'in so'zlar (madina, kamor)
Vektor farosida yag'in olma esab eganay
matn uzoqda faglastan.

3. Embedding vs Encoding

Encoding har gancha ma'lumotni barcha formatga mexanik o'girish ma'nosi jushunni bayonini.

Embedding esa natijada olingan vektor ma'nosi semantik yozilishni o'ziga saqlaydigan bir qancha maxsus kodlash turi.

4. Embedding turlari.

Asosiy 2 turi token embeddinglari (söz/token ma'nosi ni kodlaydi) va position embeddinglar (tokenning gapdagi o'rnini kodlaydi).

5. Position Embeddings: Absolute, Relative va RoPE

Transformer tokenlar tartibini o'z ichiga olgan barchasi parallel ishlaydi, shu sababli o'zaro aloqada kodlanadi.

Absolute har pozitsiyaga qat'iy vektor

Relative tokenlar orasidagi masofa ni kodlay

RoPE pozitsiyani vektorni aylantirish orqali kodlaydi mutloq va nisbiy aylantirishni birlashtiradi.

6. Nima uchun RoPE zamonaviy modellarda

Ulama, GPT aqal dunyo usul e'tibor etilgan yashiroq umumlashadi trainda ko'rilgan usullardan barchasi ishlaydi.

7. Attention, Matematic qural

Gapdagi sozlarni bir birini ko'rishiga
imkon beradi va qaysi soz qaysi biri
uchun muhimligini aniglaydi

Maqsoad eng kuchli sozni topish emas,
sozlar o'rtasidagi g'ayblik munosabati aniqlash.

8. Attention qayerda nerga qanday Embeddinglar (token + Role) Transformer layer Attentionga unatadi.

Manosiz bog'lanmagan sozlar
vektorlar o'zgarishi natijasi bog'lanmagan soz/g'ara
tasvirlar, aytilar. Shuningdek, adim natijasi
eslab qoladigan imkon beradi.

Attention yoz Transformer yoz LLMga

9. Attention vs Embedding Ragib emas bitta jamoa

Embedding sozni manosini va o'rnini (RoP)
sonlashtirib Inputga tayyorlaydi
Attention shu vektorlarni bir biriga qarab
birgalikda gap manosini quradi
Action bo'shligi.

10. Attention turlari ~~Arxitektura~~ Attention fer
o'zgar va katta rolni o'g'ir o'ladigan
bo'lishi kerak. Har sagan o'ladigan ma'lumot
qayta o'g'ir to'g'ir so'zga shu sabab eslab
qolish kerak. Q K V

1. Bitter Lessonga kora golda ishlab chiqilgan aqilli foyeslar emas, balki mashtab va amaliy hisoblash qalaba qorinadi, Nega aynan shunday!

* Ba Qomun „algoritmalar ahamiyatsiz, fagot GPU qotirish kifoye“ degani emastligini 2012-2019 yillardagi 44x va 11x ragamlar qaralay ishlatilaydi?

* Chetlangan compute budjetingiz bo'lsa, Bitter lesson falsafasi bilan Chindilla ning optimal N/D nisbati bir biriga qaralay zid ketmasdan mos keladi.

2. Scaling law formulasi $C \approx 6ND$ token soni oshirish bilan parametri oshirishlar nega foydali bo'lishi mumkin?

* GPT-3 (175B parametri, 300B token) misolida qaralay under training yuz bergan - Chindilla talab qilgan 3.5T token misolida tushunadi.

* Llama 2 nega ataylab chindilla ni optimal nisbatdan chetlatib ko'proq token bilan o'qitilgan?

3. BF16 katta modelni o'qitishda FP16 dan nega xavfli? Aynan qaysi sonli xususiyat (dynamic range yoki precision) kuring sababchasi?

* Agar hamma narsani FP32 sarg'lasangiz
terlik va xotira nuqtai nazaridolan nishoni
yozotassiz?

* Nega feed forward matnual BF16 da boyonib
veradigc lecer attention ichidagi fast max
haran FP32 da hisoblanadi?

4. GNT qayrdan xelis chiqar - forvard / 2MT
Backward 4MT qismlarini alohida alohida
hisoblab tushuntirig.

* MFU past chiqar masalas 20% ki GNT
formulas notogri xamini bilibladim.
Yaxi boshqa narsalar delolat beradi.

* Adam W nega har bir parameter aynan
16 boyt band qiladi Bu sanini 4 to
tarcibig qisimga ayatib berig.

5. Transformerda Position Embedding masalan RoK
nega alohida kerak. Attention mexanizmining
Ozi soz tartibini nega bilangchi?

* MHA, CQA, MRA orasidagi farqlar nimga bogliq
Real maxsulot qayssi birini tanlagan bolardir

* Sparse Attention va Sliding Attention ikkalas
ham hisoblashni tezlashtirish masalalar
bolisada ular tub jihatlari qanday farq
qiladi.

Men AI o'rganaman.

- ↓
- 1) Bōlaklarga bōlish
Men AI o'gana man E
M
B
 - 2) Razanga Aylantirish
Men $\rightarrow [0.2, -0.5, 0.1 \dots]$ E
D
D
AI $\rightarrow [0.9, 0.3, -0.2 \dots]$ I
n
g
 - 3) ōrnini eslab qolish
Qaysi sōz nechindidā terjari han
Razanga qōshiladi

↓

4) Sōzlar Birbiriga qaraydi
Har sōz sōzaydi menga alaqadā sōz qargir?
o'rganaman $\rightarrow$ AI ga ko'proq qaraydi.

5) Yargi mana Tōjirga Razanlar.
Ende har sōzniq razari bo'g'liq sōzlarni
han ichiga oladi.

Attention Qism

Model butun gapni fushunadi.

llama-3.3-nemotron-super-49b-v1.5

build.nvidia.com/nvidia/llama-3.3-nemotron-super-49b-v1.5/playground

ExploreModelsSkillsBlueprintsGPUDocs

nvidia/llama-3.3-nemotron-super-49b-v1.5

BuildPlaygroundModel Card

AI models generate responses and outputs based on complex algorithms and machine learning techniques, and those responses or outputs may be inaccurate, harmful, biased or indecent. By testing this model, you assume the risk of any harm caused by any response or output of the model. Please do not upload any confidential information.

"If artificial intelligence becomes smarter than humans in the future, would that be a good thing or a dangerous one? Explain briefly, with your reasoning."

Reasoning Complete

The prospect of artificial intelligence (AI) surpassing human intelligence is a double-edged sword, presenting both immense opportunities and significant risks.

Potential Benefits:

- Problem-Solving:** A superintelligent AI could accelerate solutions to complex global challenges like disease, climate change, and poverty by analyzing data and simulating outcomes far beyond human capacity.
- Efficiency and Innovation:** It could revolutionize industries (e.g., healthcare, energy, education) through optimized systems, personalized services, and breakthrough technologies, improving quality of life.
- Scientific Advancement:** AI might drive discoveries in physics, medicine, and space exploration, addressing questions that currently elude humans.

Risks and Dangers:

- Loss of Control:** If AI goals misalign with human values, even unintentionally, it could act in harmful or unpredictable ways (e.g., prioritizing efficiency over ethics).
- Existential Threats:** Poorly designed or uncontrolled AI might pose existential risks, such as resource depletion for its objectives or enabling autonomous weapons.
- Ethical and Social Issues:** Job displacement, concentration of power in few hands, and biased decision-making could exacerbate inequality and social unrest.

Conclusion: Whether this scenario is "good" or "dangerous" hinges on how humanity develops and governs AI. Proactive measures—like robust safety protocols, ethical frameworks, and international cooperation—are critical to maximizing benefits while mitigating risks. Without careful stewardship, the dangers could outweigh the advantages.

OFFONReasoning

Parameters

Refine your idea or start something new.

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Your API Rate LimitUp to 40 rpm

Chonnam National University

API Keys

Integrations

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